# Computer Networks

Assignment 1

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## Part 1.

1. .

   IP address = 10.5.16.38
   Subnet Mask = 255.255.255.0
   Network ID = 10.5.16.0

```
user@user-Veriton-S2690G-D22E2:~$ ifconfig
enp2s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.5.16.38  netmask 255.255.255.0  broadcast 10.5.16.255
        inet6 fe80::8aae:ddff:fe33:a0b6  prefixlen 64  scopeid 0x20<link>
        ether 88:ae:dd:33:a0:b6  txqueuelen 1000  (Ethernet)
        RX packets 53435  bytes 9790272 (9.7 MB)
        RX errors 0  dropped 345  overruns 0  frame 0
        TX packets 1265  bytes 130622 (130.6 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 1836  bytes 572857 (572.8 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1836  bytes 572857 (572.8 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

2. .

   We can see that for different CDN servers there are different IP addresses for the same domain. This is done for various reasons such as load balancing, lowering geographical latency. This technique allows the request to be routed to the closest server hosting the given website.

```
user@user-Veriton-S2690G-D22E2:~$ nslookup www.facebook.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
www.facebook.com        canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 157.240.13.35
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f188:84:face:b00c:0:25de

user@user-Veriton-S2690G-D22E2:~$ nslookup www.facebook.com 172.16.1.164
Server:         172.16.1.164
Address:        172.16.1.164#53

Non-authoritative answer:
www.facebook.com        canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 157.240.16.35
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f188:84:face:b00c:0:25de

user@user-Veriton-S2690G-D22E2:~$ nslookup www.facebook.com 172.16.1.165
Server:         172.16.1.165
Address:        172.16.1.165#53

Non-authoritative answer:
www.facebook.com        canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 31.13.79.35
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f12f:183:face:b00c:0:25de

user@user-Veriton-S2690G-D22E2:~$ nslookup www.facebook.com 172.16.1.166
Server:         172.16.1.166
Address:        172.16.1.166#53

Non-authoritative answer:
www.facebook.com        canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 31.13.79.35
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f12f:183:face:b00c:0:25de

user@user-Veriton-S2690G-D22E2:~$ nslookup www.facebook.com 172.16.1.180
Server:         172.16.1.180
Address:        172.16.1.180#53

Non-authoritative answer:
www.facebook.com        canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 57.144.124.1
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f33e:8:face:b00c:0:25de
```

3. .

In the following screenshot, minimum RTT, average RTT, maximum RTT and packet loss can be seen for different packet sizes. The packets have additional 28 bytes of headers included over their base size.

```
user@user-Veriton-S2690G-D22E2:~$ ping -c 10 -s 64 -W 100 10.5.16.37
PING 10.5.16.37 (10.5.16.37) 64(92) bytes of data.
72 bytes from 10.5.16.37: icmp_seq=1 ttl=64 time=0.366 ms
72 bytes from 10.5.16.37: icmp_seq=2 ttl=64 time=0.355 ms
72 bytes from 10.5.16.37: icmp_seq=3 ttl=64 time=0.351 ms
72 bytes from 10.5.16.37: icmp_seq=4 ttl=64 time=0.322 ms
72 bytes from 10.5.16.37: icmp_seq=5 ttl=64 time=0.480 ms
72 bytes from 10.5.16.37: icmp_seq=6 ttl=64 time=0.490 ms
72 bytes from 10.5.16.37: icmp_seq=7 ttl=64 time=0.488 ms
72 bytes from 10.5.16.37: icmp_seq=8 ttl=64 time=0.487 ms
72 bytes from 10.5.16.37: icmp_seq=9 ttl=64 time=0.358 ms
72 bytes from 10.5.16.37: icmp_seq=10 ttl=64 time=0.349 ms

--- 10.5.16.37 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9223ms
rtt min/avg/max/mdev = 0.322/0.404/0.490/0.067 ms
user@user-Veriton-S2690G-D22E2:~$ ping -c 10 -s 128 -W 100 10.5.16.37
PING 10.5.16.37 (10.5.16.37) 128(156) bytes of data.
136 bytes from 10.5.16.37: icmp_seq=1 ttl=64 time=0.212 ms
136 bytes from 10.5.16.37: icmp_seq=2 ttl=64 time=0.378 ms
136 bytes from 10.5.16.37: icmp_seq=3 ttl=64 time=0.370 ms
136 bytes from 10.5.16.37: icmp_seq=4 ttl=64 time=0.364 ms
136 bytes from 10.5.16.37: icmp_seq=5 ttl=64 time=0.491 ms
136 bytes from 10.5.16.37: icmp_seq=6 ttl=64 time=0.300 ms
136 bytes from 10.5.16.37: icmp_seq=7 ttl=64 time=0.480 ms
136 bytes from 10.5.16.37: icmp_seq=8 ttl=64 time=0.504 ms
136 bytes from 10.5.16.37: icmp_seq=9 ttl=64 time=0.488 ms
136 bytes from 10.5.16.37: icmp_seq=10 ttl=64 time=0.467 ms

--- 10.5.16.37 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9224ms
rtt min/avg/max/mdev = 0.212/0.405/0.504/0.092 ms
user@user-Veriton-S2690G-D22E2:~$ ping -c 10 -s 512 -W 100 10.5.16.37
PING 10.5.16.37 (10.5.16.37) 512(540) bytes of data.
520 bytes from 10.5.16.37: icmp_seq=1 ttl=64 time=0.252 ms
520 bytes from 10.5.16.37: icmp_seq=2 ttl=64 time=0.388 ms
520 bytes from 10.5.16.37: icmp_seq=3 ttl=64 time=0.372 ms
520 bytes from 10.5.16.37: icmp_seq=4 ttl=64 time=0.375 ms
520 bytes from 10.5.16.37: icmp_seq=5 ttl=64 time=0.387 ms
520 bytes from 10.5.16.37: icmp_seq=6 ttl=64 time=0.394 ms
520 bytes from 10.5.16.37: icmp_seq=7 ttl=64 time=0.379 ms
520 bytes from 10.5.16.37: icmp_seq=8 ttl=64 time=0.388 ms
520 bytes from 10.5.16.37: icmp_seq=9 ttl=64 time=0.372 ms
520 bytes from 10.5.16.37: icmp_seq=10 ttl=64 time=0.391 ms

--- 10.5.16.37 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9198ms
rtt min/avg/max/mdev = 0.252/0.369/0.394/0.040 ms
```

4. .

Number of hops - **12**
Number of visible hosts - **13** (upper bound of 12 * 3 = 36)

In some hops * * * is seen, which represents that the intermediate host did not respond. This can be caused due to multiple reasons such as router overload, firewall, port blocking

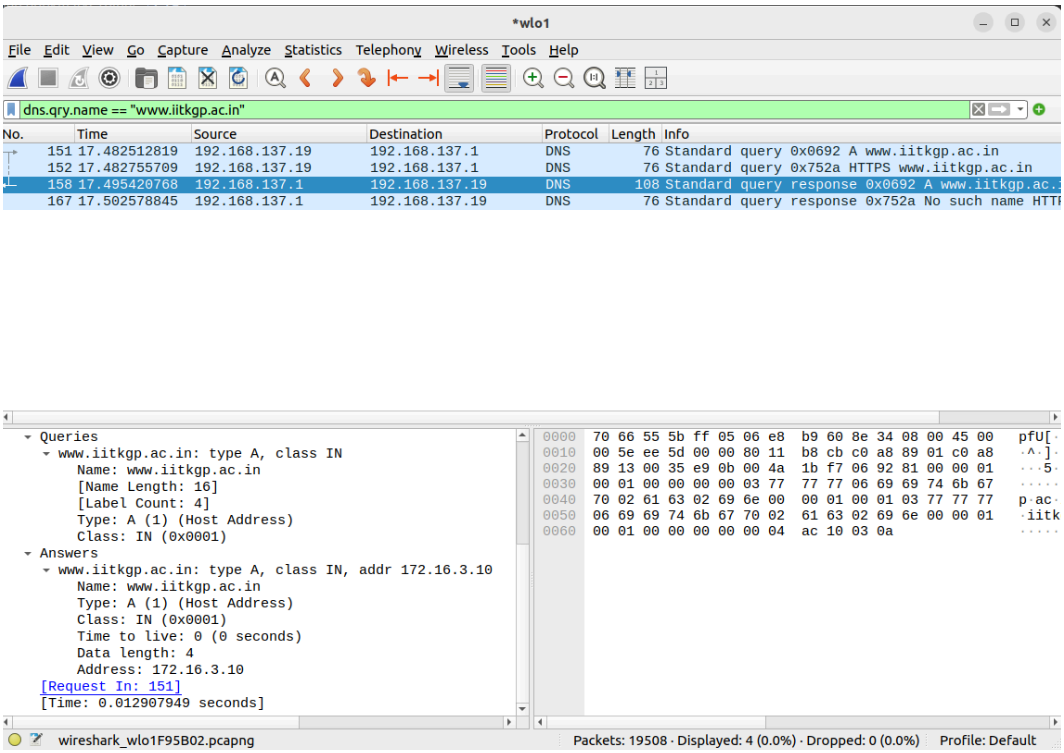
```
user@user-Veriton-S2690G-D22E2:~$ traceroute www.google.com
traceroute to www.google.com (142.250.183.164), 30 hops max, 60 byte packets
 1  _gateway (10.5.16.2)  0.518 ms  0.483 ms  0.470 ms
 2  10.120.2.33 (10.120.2.33)  0.457 ms  0.443 ms  0.431 ms
 3  10.255.1.3 (10.255.1.3)  3.828 ms  3.828 ms  3.920 ms
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  72.14.204.62 (72.14.204.62)  37.892 ms  46.059 ms  46.547 ms
 9  * * *
10  142.250.214.102 (142.250.214.102)  56.496 ms 142.251.77.100 (142.251.77.100)  39.879 ms 172.253.77.22 (172.253.77.22)  40.906 ms
11  142.251.64.15 (142.251.64.15)  46.545 ms 142.251.64.13 (142.251.64.13)  46.456 ms 142.251.64.15 (142.251.64.15)  49.012 ms
12  142.250.208.227 (142.250.208.227)  58.569 ms bom07s32-in-f4.1e100.net (142.250.183.164)  46.782 ms 142.250.226.67 (142.250.226.67)  46.664 ms
```

# Part 2.

1. Analysis of DNS Packets: Structure and its Traffic
    a. DNS uses UDP for packet transmission
    b. IP of source machine - 192.168.137.19, IP of destination machine - 192.168.137.1
    c. 2 DNS queries were sent to the DNS server
    d. 1 server responded with actual IP address in query for A type record.
    e. Only 1 DNS server was involved which responded.

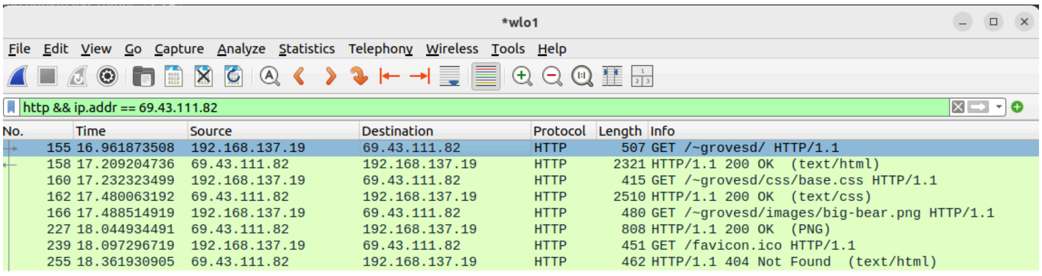
```
parth@parth-HP-Laptop-15s-fr1xxx:~$ ( nmcli dev list || nmcli dev show ) 2>/dev/null | grep DNS
IP4.DNS[1]:                             192.168.137.1
```

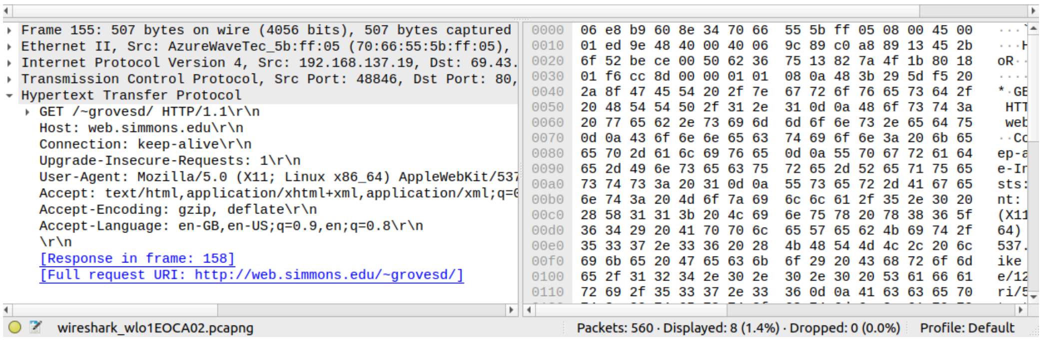
    f. .



2. Web Traffic (HTTP)

a. .




b. Header of requests and responses have different format which can be analyzed to see which packet belongs to request and which to response
HTTP request headers contain fields like Request Method, Request URI, Request Version, Host, User-Agent, Referrer etc.
HTTP response headers have fields like Response Version, Status Code, Response Phrase, Date, Content-Type etc.

c. 8 packets are exchanged between client (browser) and server (69.43.111.82) to load the entire web page

3. ICMP Traffic (Ping/Traceroute)

a. .

Source of IP packets - 10.145.156.238

Destination of packets - 10.145.22.3

```
parth@parth-HP-Laptop-15s-fr1xxx:~$ ping 10.145.22.3
PING 10.145.22.3 (10.145.22.3) 56(84) bytes of data.
64 bytes from 10.145.22.3: icmp_seq=1 ttl=61 time=319 ms
64 bytes from 10.145.22.3: icmp_seq=2 ttl=61 time=148 ms
64 bytes from 10.145.22.3: icmp_seq=3 ttl=61 time=161 ms
64 bytes from 10.145.22.3: icmp_seq=4 ttl=61 time=184 ms
64 bytes from 10.145.22.3: icmp_seq=5 ttl=61 time=109 ms
64 bytes from 10.145.22.3: icmp_seq=6 ttl=61 time=244 ms
64 bytes from 10.145.22.3: icmp_seq=7 ttl=61 time=455 ms
64 bytes from 10.145.22.3: icmp_seq=8 ttl=61 time=180 ms
64 bytes from 10.145.22.3: icmp_seq=9 ttl=61 time=195 ms
64 bytes from 10.145.22.3: icmp_seq=10 ttl=61 time=217 ms
64 bytes from 10.145.22.3: icmp_seq=11 ttl=61 time=447 ms
^C
--- 10.145.22.3 ping statistics ---
12 packets transmitted, 11 received, 8.33333% packet loss, time 11011ms
rtt min/avg/max/mdev = 109.063/241.787/455.192/111.360 ms
```

| | | | | | | |
|---|---|---|---|---|---|---|
| 3 | 9.611706 | 10.145.156.238 | 10.145.22.3 | ICMP | 98 | Echo (ping) request |
| 4 | 9.930639 | 10.145.22.3 | 10.145.156.238 | ICMP | 98 | Echo (ping) reply |
| 5 | 10.612696 | 10.145.156.238 | 10.145.22.3 | ICMP | 98 | Echo (ping) request |
| 6 | 10.761106 | 10.145.22.3 | 10.145.156.238 | ICMP | 98 | Echo (ping) reply |
| 7 | 11.614328 | 10.145.156.238 | 10.145.22.3 | ICMP | 98 | Echo (ping) request |
| 8 | 11.775514 | 10.145.22.3 | 10.145.156.238 | ICMP | 98 | Echo (ping) reply |
| 9 | 12.615729 | 10.145.156.238 | 10.145.22.3 | ICMP | 98 | Echo (ping) request |
| 10 | 12.799641 | 10.145.22.3 | 10.145.156.238 | ICMP | 98 | Echo (ping) reply |
| 11 | 13.616822 | 10.145.156.238 | 10.145.22.3 | ICMP | 98 | Echo (ping) request |
| 12 | 13.725853 | 10.145.22.3 | 10.145.156.238 | ICMP | 98 | Echo (ping) reply |
| 13 | 14.618122 | 10.145.156.238 | 10.145.22.3 | ICMP | 98 | Echo (ping) request |
| 14 | 14.849380 | AzureWaveTec_5b:ff:… | Fortinet_09:01:1e | ARP | 42 | Who has 10.145.128.2? |
| 15 | 14.861923 | 10.145.22.3 | 10.145.156.238 | ICMP | 98 | Echo (ping) reply |
| 16 | 14.861923 | Fortinet_09:01:1e | AzureWaveTec_5b:ff:… | ARP | 56 | 10.145.128.2 is at 00 |
| 17 | 15.619451 | 10.145.156.238 | 10.145.22.3 | ICMP | 98 | Echo (ping) request |
| 18 | 16.074603 | 10.145.22.3 | 10.145.156.238 | ICMP | 98 | Echo (ping) reply |
| 19 | 16.619513 | 10.145.156.238 | 10.145.22.3 | ICMP | 98 | Echo (ping) request |
| 20 | 16.799165 | 10.145.22.3 | 10.145.156.238 | ICMP | 98 | Echo (ping) reply |
| 21 | 17.621343 | 10.145.156.238 | 10.145.22.3 | ICMP | 98 | Echo (ping) request |
| 22 | 17.816382 | 10.145.22.3 | 10.145.156.238 | ICMP | 98 | Echo (ping) reply |
| 23 | 18.621776 | 10.145.156.238 | 10.145.22.3 | ICMP | 98 | Echo (ping) request |

3 packets have TTL = 1 and 3 have TTL = 2, which is the expected behaviour according to tracerouting algorithm.

```
parth@parth-HP-Laptop-15s-fr1xxx:~$ traceroute 10.145.22.3
traceroute to 10.145.22.3 (10.145.22.3), 30 hops max, 60 byte packets
 1  _gateway (10.145.128.2)  11.581 ms  11.516 ms  11.490 ms
 2  10.250.1.4 (10.250.1.4)  11.473 ms  11.456 ms  11.440 ms
 3  10.120.0.26 (10.120.0.26)  11.421 ms  11.753 ms  11.388 ms
 4  10.145.22.3 (10.145.22.3)  494.283 ms  494.266 ms  494.249 ms
```

| | | | | |
|---|---|---|---|---|
| 31 30.837534 | 10.145.156.238 | 10.145.22.3 | UDP | 74 58007 → 33434 Len=32 |
| 32 30.837573 | 10.145.156.238 | 10.145.22.3 | UDP | 74 37319 → 33435 Len=32 |
| 33 30.837598 | 10.145.156.238 | 10.145.22.3 | UDP | 74 38176 → 33436 Len=32 |
| 34 30.837615 | 10.145.156.238 | 10.145.22.3 | UDP | 74 53682 → 33437 Len=32 |
| 35 30.837632 | 10.145.156.238 | 10.145.22.3 | UDP | 74 38247 → 33438 Len=32 |
| 36 30.837649 | 10.145.156.238 | 10.145.22.3 | UDP | 74 36148 → 33439 Len=32 |
| 37 30.837666 | 10.145.156.238 | 10.145.22.3 | UDP | 74 53858 → 33440 Len=32 |
| 38 30.837682 | 10.145.156.238 | 10.145.22.3 | UDP | 74 59277 → 33441 Len=32 |
| 39 30.837699 | 10.145.156.238 | 10.145.22.3 | UDP | 74 49703 → 33442 Len=32 |
| 40 30.837717 | 10.145.156.238 | 10.145.22.3 | UDP | 74 39663 → 33443 Len=32 |
| 41 30.837733 | 10.145.156.238 | 10.145.22.3 | UDP | 74 55777 → 33444 Len=32 |
| 42 30.837750 | 10.145.156.238 | 10.145.22.3 | UDP | 74 41254 → 33445 Len=32 |
| 43 30.837769 | 10.145.156.238 | 10.145.22.3 | UDP | 74 36949 → 33446 Len=32 |
| 44 30.837786 | 10.145.156.238 | 10.145.22.3 | UDP | 74 58464 → 33447 Len=32 |
| 45 30.837803 | 10.145.156.238 | 10.145.22.3 | UDP | 74 35762 → 33448 Len=32 |
| 46 30.837820 | 10.145.156.238 | 10.145.22.3 | UDP | 74 56350 → 33449 Len=32 |
| 47 30.849083 | 10.145.128.2 | 10.145.156.238 | ICMP | 102 Time-to-live exceeded |
| 48 30.849083 | 10.145.128.2 | 10.145.156.238 | ICMP | 102 Time-to-live exceeded |
| 49 30.849083 | 10.145.128.2 | 10.145.156.238 | ICMP | 102 Time-to-live exceeded |
| 50 30.849083 | 10.120.0.26 | 10.145.156.238 | ICMP | 70 Time-to-live exceeded |
| 51 30.849083 | 10.120.0.26 | 10.145.156.238 | ICMP | 70 Time-to-live exceeded |
| 52 30.849084 | 10.250.1.4 | 10.145.156.238 | ICMP | 70 Time-to-live exceeded |
| 53 30.849084 | 10.250.1.4 | 10.145.156.238 | ICMP | 70 Time-to-live exceeded |
| 54 30.849084 | 10.250.1.4 | 10.145.156.238 | ICMP | 70 Time-to-live exceeded |
| 55 30.849431 | 10.120.0.26 | 10.145.156.238 | ICMP | 70 Time-to-live exceeded |

b. .

Sent ICMP packets receive no response.

```
parth@parth-HP-Laptop-15s-fr1xxx:~$ ping 192.168.31.3
PING 192.168.31.3 (192.168.31.3) 56(84) bytes of data.
^C
--- 192.168.31.3 ping statistics ---
11 packets transmitted, 0 received, 100% packet loss, time 10278ms
```

| | | | | |
|---|---|---|---|---|
| 91 60.314177 | 10.145.156.238 | 192.168.31.3 | ICMP | 98 Echo (ping) request |
| 92 61.376616 | 10.145.156.238 | 192.168.31.3 | ICMP | 98 Echo (ping) request |
| 93 62.400615 | 10.145.156.238 | 192.168.31.3 | ICMP | 98 Echo (ping) request |
| 94 63.424529 | 10.145.156.238 | 192.168.31.3 | ICMP | 98 Echo (ping) request |
| 95 64.448591 | 10.145.156.238 | 192.168.31.3 | ICMP | 98 Echo (ping) request |
| 98 65.472524 | 10.145.156.238 | 192.168.31.3 | ICMP | 98 Echo (ping) request |
| 101 66.496581 | 10.145.156.238 | 192.168.31.3 | ICMP | 98 Echo (ping) request |
| 102 67.520583 | 10.145.156.238 | 192.168.31.3 | ICMP | 98 Echo (ping) request |
| 103 68.544588 | 10.145.156.238 | 192.168.31.3 | ICMP | 98 Echo (ping) request |
| 104 69.568591 | 10.145.156.238 | 192.168.31.3 | ICMP | 98 Echo (ping) request |
| 105 70.592597 | 10.145.156.238 | 192.168.31.3 | ICMP | 98 Echo (ping) request |

c. .

While trying to trace an unreachable host, the first 6 hops are being traced and a packet is being returned. But after that there is no packet reply, hence it can be assumed that there is no viable route to reach the destination address. (Default max. number of hops is 30)

```
parth@parth-HP-Laptop-15s-fr1xxx:~$ traceroute 192.168.31.3
traceroute to 192.168.31.3 (192.168.31.3), 30 hops max, 60 byte packets
 1  _gateway (10.145.128.2)  9.714 ms  9.645 ms  8.560 ms
 2  192.168.255.18 (192.168.255.18)  8.538 ms  8.521 ms  8.862 ms
 3  10.119.228.129 (10.119.228.129)  8.476 ms  8.460 ms  8.443 ms
 4  10.173.35.1 (10.173.35.1)  38.255 ms 10.173.35.253 (10.173.35.253)  107.832 ms  107.810 ms
 5  10.255.239.189 (10.255.239.189)  107.794 ms 10.255.238.166 (10.255.238.166)  38.183 ms 10.255.239.189 (10.255.239.189)  107.760 ms
 6  10.152.7.38 (10.152.7.38)  107.741 ms  93.061 ms  92.993 ms
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```